

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	138781	46.074805
1	[Git, Change, Create]	44929	14.916270
2	[Git, Change, Delete]	22781	7.563212
3	[Java, ByteCode, Method, Member]	13517	4.487597
4	[Java, ByteCode, Parameter]	13306	4.417545
5	[Git, Commit]	11390	3.781440
6	[File, Git]	8286	2.750923
7	[Java, ByteCode, Bound, ParameterizedType]	7286	2.418926
8	[Java, ByteCode, Bound]	7245	2.405315
9	[Git, Change, Rename]	7018	2.329951
10	[Java, ByteCode, Member, Field]	3606	1.197179
11	[Java, ByteCode, Bound, WildcardType]	2950	0.979390
12	[Java, Value, ByteCode, Annotation]	2936	0.974742
13	[Xml, Element]	2162	0.717776
14	[Java, ByteCode, Constructor, Method, Member]	2109	0.700181
15	[Xml, Text]	1450	0.481395
16	[Java, ByteCode, TypeVariable, Bound]	1111	0.368848
17	[Java, ByteCode, Method, Member, Lambda]	968	0.321373
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	890	0.295477
19	[Type, File, Java, ByteCode, Class]	845	0.280537
20	[Json, Key]	739	0.245345
21	[Value, Json, Scalar]	723	0.240033
22	[Java, Value, ByteCode, Primitive]	673	0.223434
23	[Type, File, Java, ByteCode, JavaType]	649	0.215466
24	[Java, ByteCode, Method, Member, GenericDeclar...	578	0.191894
25	[Type, File, Java, ByteCode, ExternalType]	412	0.136783
26	[Git, Change, Copy]	298	0.098935
27	[Author, Git, Person]	295	0.097939
28	[Value, Array]	289	0.095947
29	[Committer, Git, Person]	242	0.080343

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

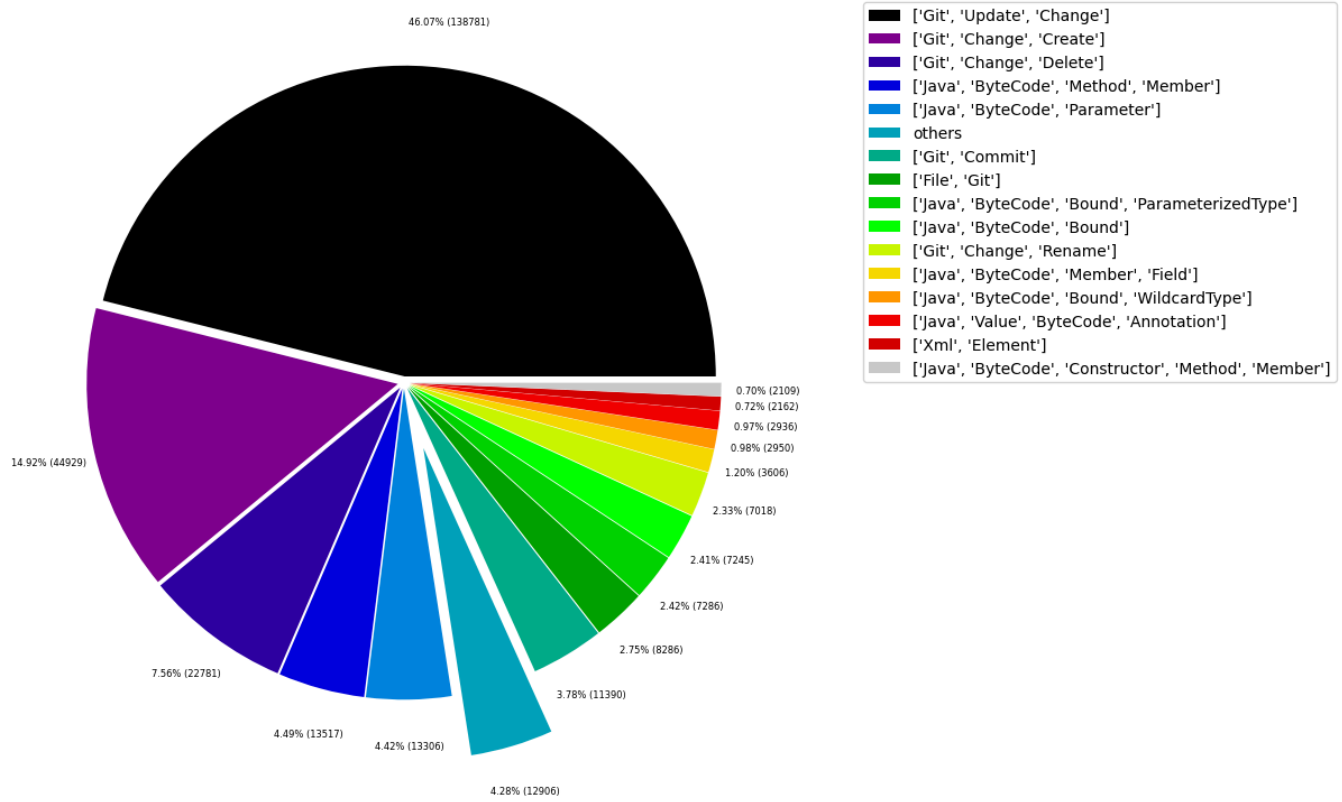


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000332
1	[Git, Branch]	1	0.000332
2	[Repository, File, Git]	1	0.000332
3	[File, Json]	2	0.000664
4	[File]	3	0.000996
5	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001328
6	[Maven, Exclusion]	5	0.001660
7	[Value, Array, Json]	6	0.001992
8	[Type, File, Java, ByteCode, Void]	9	0.002988
9	[File, Maven, Xml, Pom, Document]	9	0.002988
10	[Java, ManifestSection]	9	0.002988
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002988
12	[File, Java, Manifest]	9	0.002988
13	[File, Java, ServiceLoader]	10	0.003320
14	[File, Java, Properties]	12	0.003984
15	[Maven, ExecutionGoal]	16	0.005312
16	[Maven, PluginExecution]	16	0.005312
17	[Xml, Attribute]	18	0.005976
18	[JQAssistant, Rule, Concept]	19	0.006308
19	[Maven, Plugin]	21	0.006972
20	[Maven, Configuration]	21	0.006972
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006972
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007304
23	[Type, File, Java, ByteCode, Enum]	28	0.009296
24	[Type, File, Java, ByteCode, PrimitiveType]	30	0.009960
25	[Xml, Namespace]	36	0.011952
26	[Type, File, Java, ByteCode, Annotation]	44	0.014608
27	[File, Directory]	44	0.014608
28	[Type, File, Java, ByteCode, Class, Throwable]	55	0.018260
29	[Type, File, Java, ByteCode, GenericDeclaratio...	85	0.028220

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

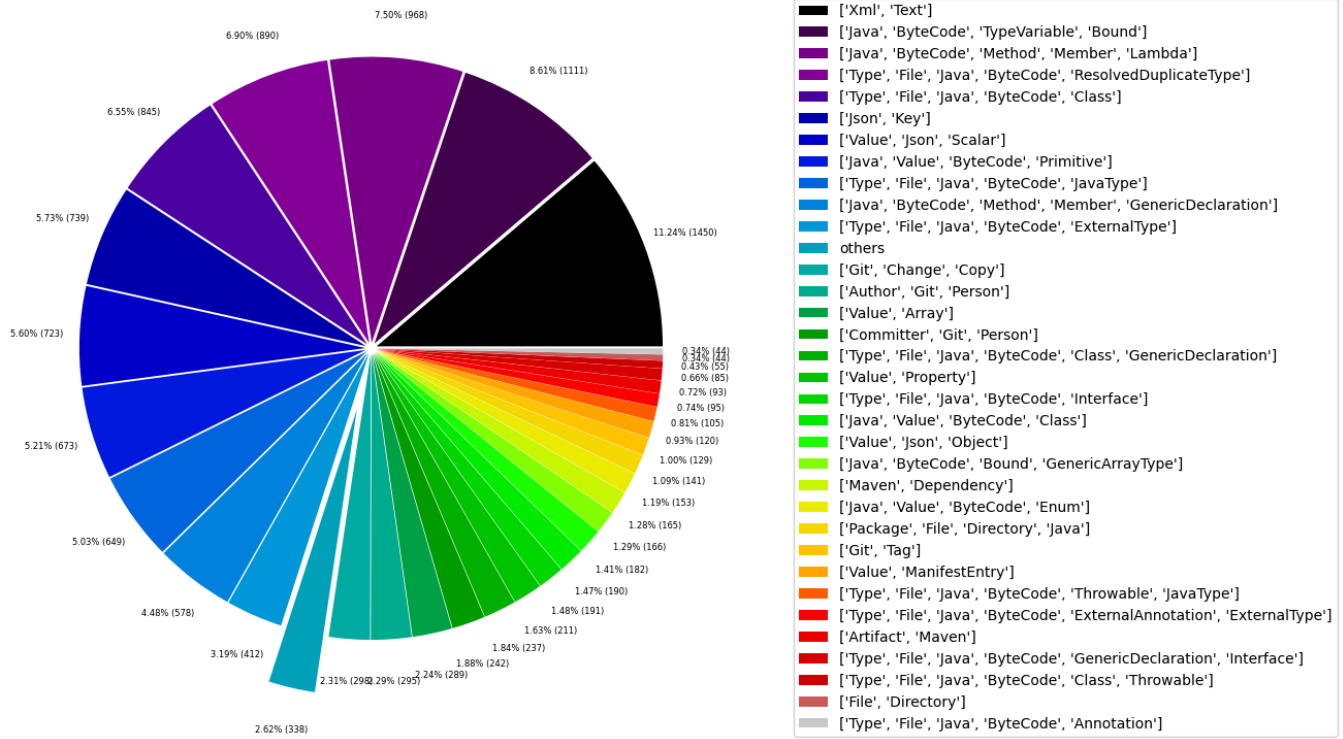


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

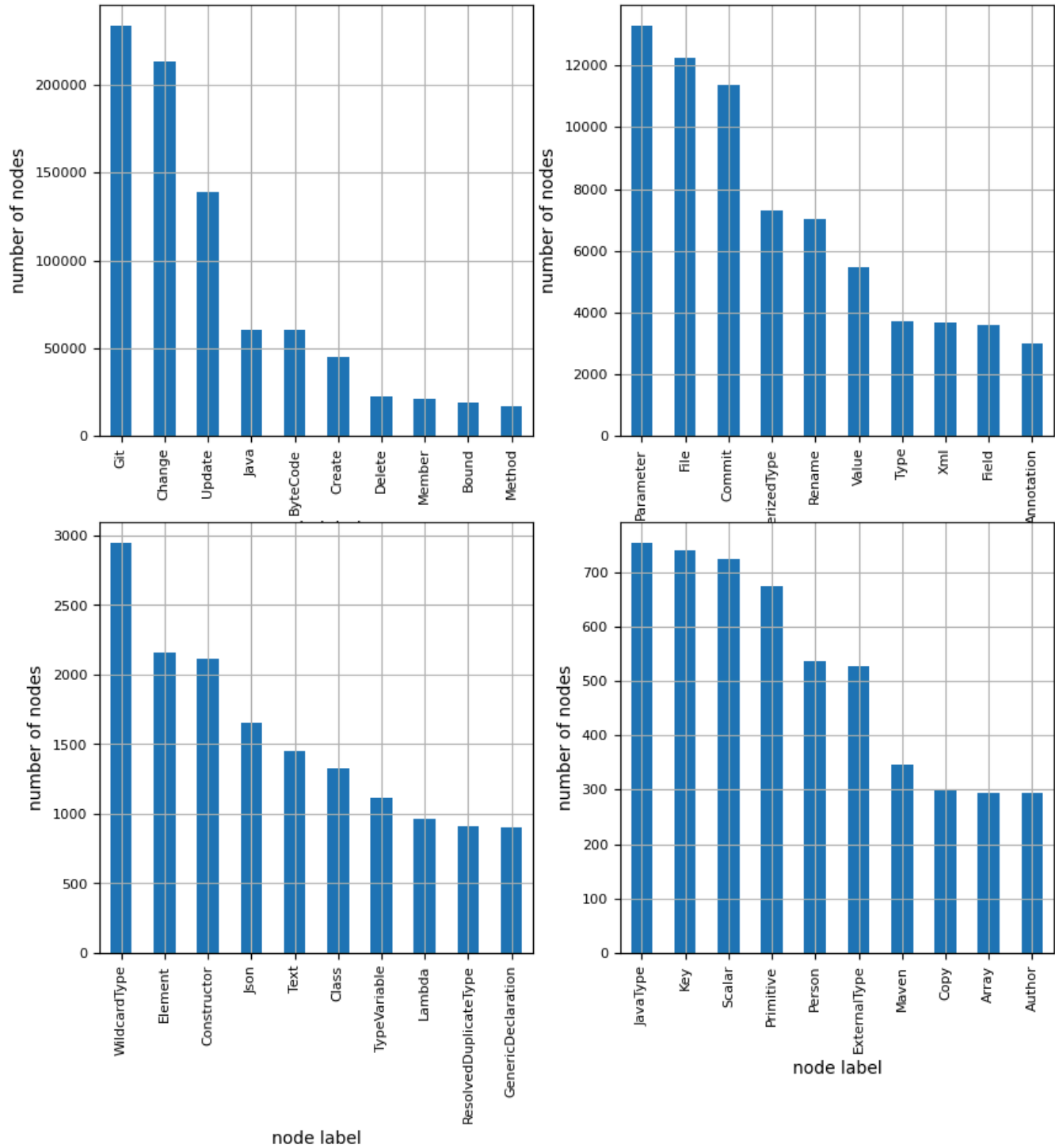
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	234151	77.737311
1	Change	213807	70.983174
2	Update	138781	46.074805
3	Java	60706	20.154179
4	ByteCode	60516	20.091100
5	Create	44929	14.916270
6	Delete	22781	7.563212
7	Member	20782	6.899551
8	Bound	18758	6.227590
9	Method	17176	5.702372
10	Parameter	13306	4.417545
11	File	12244	4.064965
12	Commit	11390	3.781440
13	ParameterizedType	7286	2.418926
14	Rename	7018	2.329951
15	Value	5483	1.820337
16	Type	3718	1.234363
17	Xml	3675	1.220087
18	Field	3606	1.197179
19	Annotation	2980	0.989350
20	WildcardType	2950	0.979390
21	Element	2162	0.717776
22	Constructor	2113	0.701509
23	Json	1652	0.548458
24	Text	1450	0.481395
25	Class	1327	0.440559
26	TypeVariable	1111	0.368848
27	Lambda	968	0.321373
28	ResolvedDuplicateType	912	0.302781
29	GenericDeclaration	904	0.300125
30	JavaType	754	0.250325
31	Key	739	0.245345
32	Scalar	723	0.240033
33	Primitive	673	0.223434
34	Person	537	0.178282
35	ExternalType	528	0.175294
36	Maven	346	0.114871
37	Copy	298	0.098935
38	Array	295	0.097939
39	Author	295	0.097939

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 935970

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	213807	22.843360
1	MODIFIES	213807	22.843360
2	UPDATES	138781	14.827505
3	CREATES	52245	5.581910
4	INVOKES	36640	3.914655
5	DELETES	29799	3.183756
6	COMMITTED	22780	2.433839
7	DEPENDS_ON	22504	2.404351
8	OF_TYPE	21945	2.344626
9	DECLARES	21242	2.269517
10	OF_RAW_TYPE	17367	1.855508
11	HAS	14444	1.543212
12	HAS_PARENT	13921	1.487334
13	RETURNS	12858	1.373762
14	HAS_COMMIT	11390	1.216919
15	READS	9385	1.002703
16	HAS_ACTUAL_TYPE_ARGUMENT	8414	0.898960
17	HAS_FILE	8286	0.885285
18	RENAMES	7018	0.749810
19	OF_GENERIC_TYPE	6000	0.641046
20	RESOLVES_TO	4236	0.452579
21	SIMILAR	4078	0.435698
22	WRITES	3948	0.421808
23	CONTAINS	3928	0.419672
24	HAS_NEW_NAME	3662	0.391252
25	RETURNS_GENERIC	3597	0.384307
26	CHANGED_TOGETHER_WITH	3034	0.324156
27	ANNOTATED_BY	2924	0.312403
28	REQUIRES	2233	0.238576
29	HAS_FIRST_CHILD	2162	0.230990

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

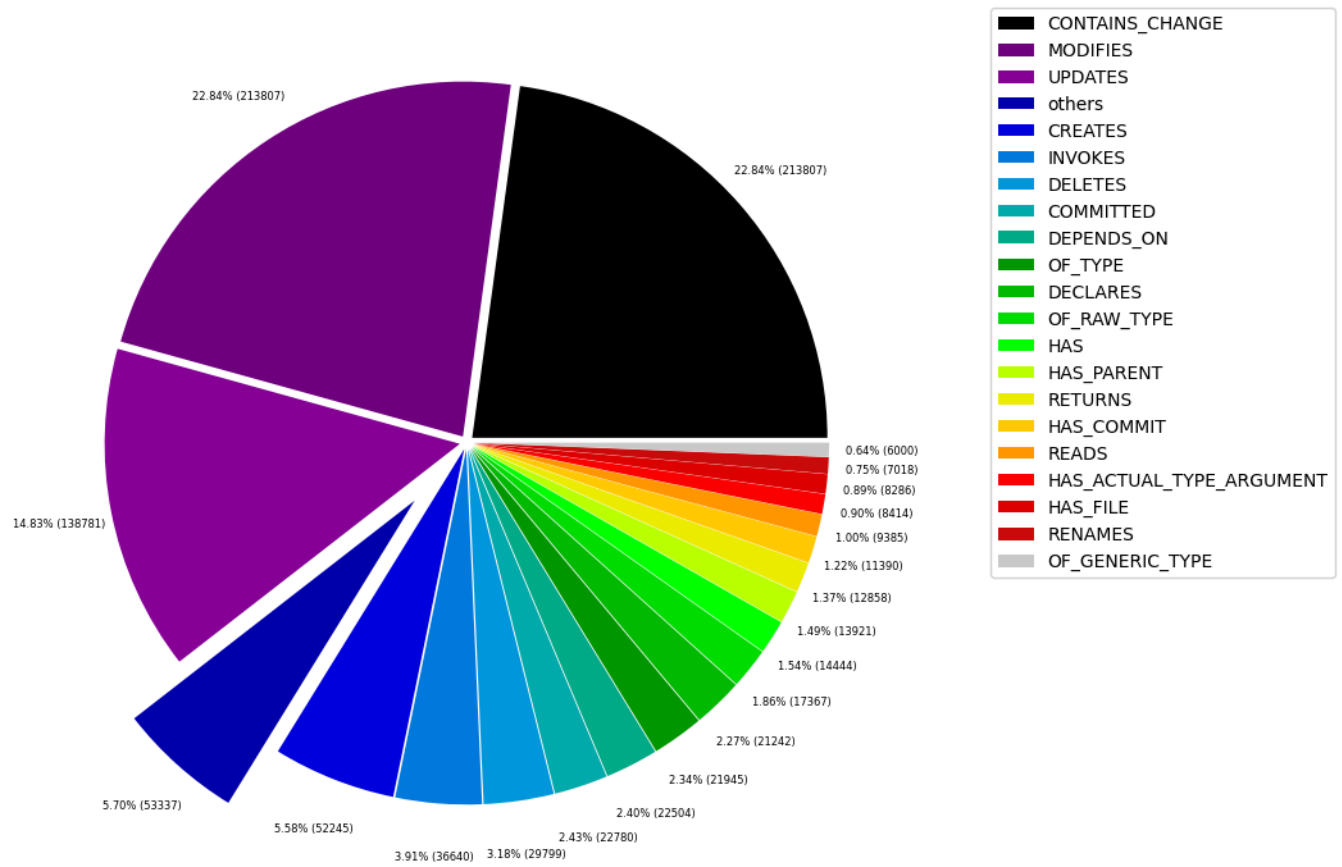


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

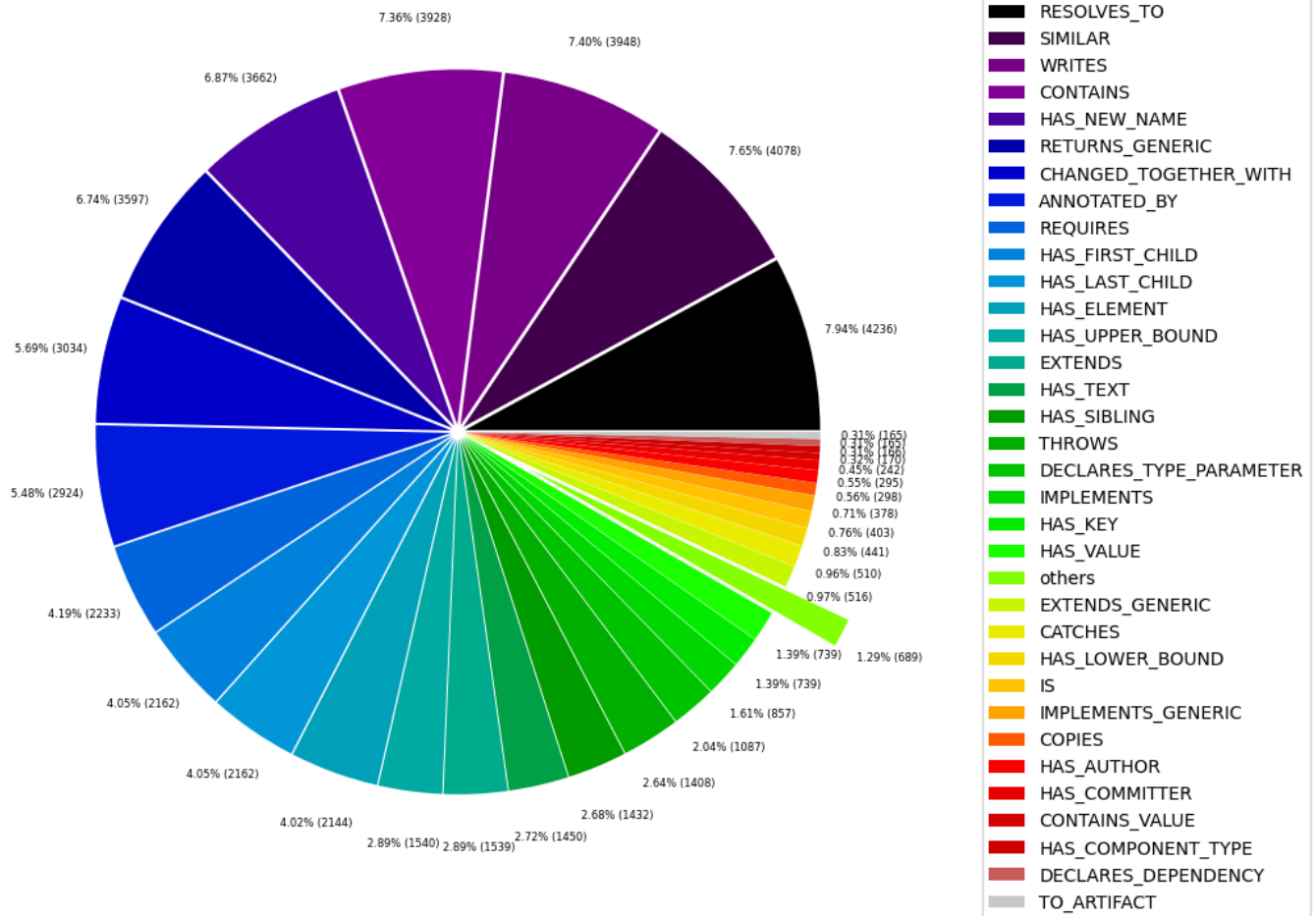
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000107
1	HAS_BRANCH	1	0.000107
2	HAS_HEAD	2	0.000214
3	THROWS_GENERIC	5	0.000534
4	EXCLUDES	5	0.000534
5	DESCRIBES	9	0.000962
6	HAS_ROOT_ELEMENT	12	0.001282
7	HAS_GOAL	16	0.001709
8	HAS_EXECUTION	16	0.001709
9	OF_NAMESPACE	18	0.001923
10	HAS_ATTRIBUTE	18	0.001923
11	INCLUDES_CONCEPT	19	0.002030
12	USES_PLUGIN	21	0.002244
13	IS_ARTIFACT	21	0.002244
14	HAS_CONFIGURATION	21	0.002244
15	REQUIRES_TYPE_PARAMETER	24	0.002564
16	REQUIRES_CONCEPT	28	0.002992
17	DECLARES_NAMESPACE	36	0.003846
18	HAS_DEFAULT	39	0.004167
19	COPY_OF	119	0.012714
20	HAS_TAG	129	0.013782
21	ON_COMMIT	129	0.013782
22	DECLARES_DEPENDENCY	165	0.017629
23	TO_ARTIFACT	165	0.017629
24	HAS_COMPONENT_TYPE	166	0.017736
25	CONTAINS_VALUE	170	0.018163
26	HAS_COMMITTER	242	0.025856
27	HAS_AUTHOR	295	0.031518
28	COPIES	298	0.031839
29	IMPLEMENTS_GENERIC	378	0.040386

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	MODIFIES	[File, Git]	138781	138781	
1	[Git, Update, Change]	UPDATES	[File, Git]	138781	138781	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	138781	11390	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	44929	11390	
4	[Git, Change, Create]	MODIFIES	[File, Git]	44929	44929	
5	[Git, Change, Create]	CREATES	[File, Git]	44929	44929	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22781	11390	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	22781	22781	
8	[Git, Change, Delete]	DELETES	[File, Git]	22781	22781	
9	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	22306	13418	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	13912	11390	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11390	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	11390	295	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11390	242	
14	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8516	13418	
15	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8430	13418	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8286	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11390	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	6160	13306	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8286	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	3466	7245	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	3114	7286	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7286	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2698	13306	
28	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2520	2109	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, TypeVariable, Bound]	2475	7286	

Graph Density

```
total_number_of_nodes (vertices): 301208
total_number_of_relationships (edges): 935970
-> total directed graph density: 1.0316452094363982e-05
-> total directed graph density in percent: 0.0010316452094363982
```